

TCP/IP SUITE THEORY-

1. **Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**



Software Required

- Wireshark
- Internet connection
- Web browser

Procedure

1. Open Wireshark.
2. Select the active network interface (Wi-Fi or Ethernet).
3. Click Start Capturing Packets.
4. Open a web browser and visit www.flipkart.com.
5. Refresh the homepage once.
6. Stop the packet capture after a few seconds.
7. Examine the captured packets and identify packets from different protocol layers.

Observation

TCP/IP Layer	Protocol Found	Header Information Observed
Application Layer	HTTPS	Secure communication between browser and Flipkart server
Transport Layer	TCP	Source Port, Destination Port, Sequence Number, ACK Number
Internet Layer	IPv4	Source IP Address, Destination IP Address
Network Access Layer	Ethernet II	Source MAC Address, Destination MAC Address

Sample Packet Details

1. Application Layer (HTTPS/TLS)
 - Protocol: HTTPS (TLS)
 - Purpose: Secure transfer of web page data between browser and server.
2. Transport Layer (TCP)
 - Protocol: TCP
 - Source Port: Random Port

- Destination Port: 443
 - Purpose: Provides reliable communication.
3. Internet Layer (IPv4)
- Protocol: IPv4
 - Contains Source IP and Destination IP addresses.
 - Purpose: Routes packets between devices.
4. Network Access Layer (Ethernet II)
- Protocol: Ethernet II
 - Contains Source MAC Address and Destination MAC Address.
 - Purpose: Transfers data over the local network.

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).



Comparison of OSI Model and TCP/IP Model

OSI Model Layer	TCP/IP Model Layer	Real-World Example (WhatsApp Message)
Application	Application	User types and sends a message on WhatsApp.
Presentation	Application	WhatsApp encrypts the message before sending.
Session	Application	WhatsApp keeps the chat session active between users.
Transport	Transport	TCP ensures the message is delivered correctly without data loss.
Network	Internet	IP finds the destination address and routes the message over the Internet.
Data Link	Network Access	Wi-Fi transfers data from the laptop/mobile to the router.
Physical	Network Access	Electrical signals or radio waves carry the data through the network cable or Wi-Fi.

3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

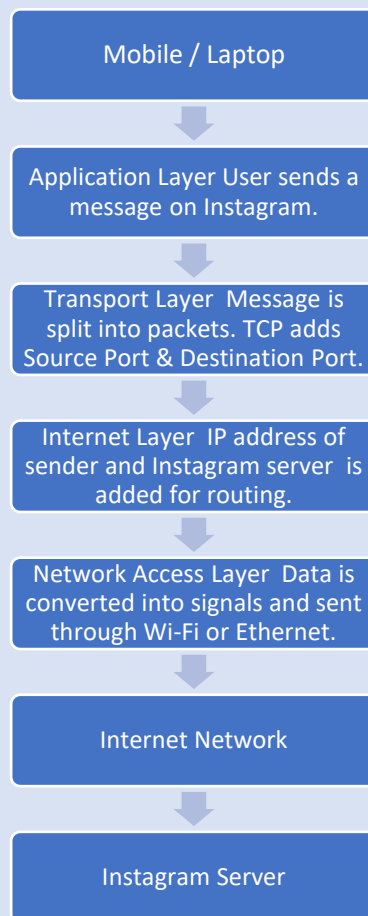


TCP Segment Header Fields and Their Functions

TCP Header Field	Function
Source Port	Identifies the port number of the sender application. It tells where the data originated.
Destination Port	Identifies the port number of the receiving application. It tells where the data should be delivered.
Sequence Number	Indicates the order of data bytes in the communication. It helps in arranging data correctly.
Acknowledgment Number	Confirms the successful receipt of data and indicates the next expected sequence number.

Header Length (Data Offset)	Specifies the size of the TCP header so the receiver knows where the actual data begins.
Flags (Control Bits)	Control the TCP connection. Common flags are SYN, ACK, FIN, RST, PSH, and URG.
Window Size	Indicates how much data the receiver can accept at one time. It helps control the flow of data.
Checksum	Detects errors in the TCP header and data during transmission.
Urgent Pointer	Points to urgent data when the URG flag is set. It is rarely used in normal communication.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.



Packet Analysis Using Wireshark

Selected Packet: HTTP Request

Packet Name	HTTP Request
TCP/IP Layer	Application Layer
How I Identified It	In Wireshark, the Protocol column showed HTTP , which indicates that it belongs to the Application Layer.
Real-World Action	This packet represents opening a webpage or sending a request from an application. For example, when I open Instagram or BookMyShow, my device sends an HTTP request to the server to load the page or app data.